

older women. The exposure/intervention (E) included HRT regimens used in postmenopausal women, including estrogen-only therapy and combined estrogen–progestogen therapy, regardless of formulation, dose, route of administration, duration of treatment, or timing of initiation. The comparator (C) included placebo, usual care, or alternative HRT regimens. The outcomes (O) of interest were cardiovascular, renal, and metabolic outcomes. Cardiovascular outcomes included major cardiovascular events and related endpoints, such as coronary heart disease, myocardial infarction, stroke, heart failure, hypertension, atrial fibrillation, venous thromboembolism, deep vein thrombosis, pulmonary embolism, and composite cardiovascular outcomes. Renal outcomes included kidney disease, chronic kidney disease, acute kidney injury, albuminuria, proteinuria, and changes in glomerular filtration rate. Metabolic outcomes included metabolic syndrome, type 2 diabetes, insulin resistance, dyslipidemia, obesity, overweight, and glycemic outcomes.

Eligible study designs included RCTs and secondary analyses of randomized trial populations, including subgroup analyses and post-trial or long-term follow-up reports. Studies were excluded if they involved premenopausal women, did not provide findings relevant to older postmenopausal women, included populations outside the review scope, evaluated interventions other than HRT, or did not report outcomes of interest. Observational designs, including prospective cohort studies, retrospective cohort studies, case-control studies, cross-sectional studies, and registry-based analyses, were excluded because HRT use in these studies may be influenced by baseline characteristics, indication for treatment, prescribing patterns, and other factors that could introduce confounding and bias the final estimates. Although the exclusion of observational studies narrowed the review to evidence arising from randomized exposure, it also prevented the review from representing the full contemporary evidence base. Additionally, reviews, editorials, commentaries, letters, protocols, preclinical studies, and conference abstracts without sufficient data were excluded. No restrictions were placed on geographic location or clinical setting, but only English-language studies were included.

Information Sources and Search Strategy

A systematic literature search was conducted in PubMed, Embase, Scopus, and the Cochrane Library, including the Cochrane Central Register of Controlled Trials (CENTRAL), to identify eligible literature. The search covered studies published from January 1, 2021, through February 28, 2026. Search strategies combined database-specific controlled vocabulary and free-text terms related to menopausal hormone therapy, including estrogen-only therapy, combined estrogen–progestogen therapy, estrogen replacement therapy, postmenopausal hormone therapy, systemic estrogen therapy, transdermal estradiol, and tibolone, together with terms for menopause and for cardiovascular, metabolic, and renal outcomes. Where available, filters for publication year, language, sex, and age were applied at the database level; otherwise, eligibility criteria were assessed during manual screening. Full search strategies for all databases are provided in the [Supplementary Material](#) and [Table S1](#).

Data Collection Process and Data Items

Data extraction was performed independently by two reviewers (TM and MFH) using a standardized data extraction form. Disagreements were resolved through discussion and consensus.

The extracted data included study characteristics such as author, publication year, study design, trial name when applicable, country or setting, and duration of follow-up; participant characteristics, including sample size, age, and menopausal status; intervention and comparator details, including HRT regimen, formulation, dose, route of administration, duration of treatment, and comparator type; and reported cardiovascular, renal, and metabolic outcomes, including outcome definitions, effect estimates, and measures of variability when available.

Risk of Bias (RoB) Assessment

RoB was assessed independently by two reviewers using the Cochrane RoB 2 tool,¹⁷ with disagreements resolved through discussion and consensus. RoB 2 was applied to assess potential bias across the relevant domains for each included study, including bias arising from the randomization process, deviations from intended interventions, missing outcome data, outcome measurement, and selection of the reported result. An overall risk-of-bias judgment was then assigned for each study outcome based on these domain-level assessments. When published reports did not provide sufficient information to support